

**The Impact of Emerging Technology on Businesses and Information
Systems: Cloud Computing**

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Introduction

Cloud computing has transformed how companies oversee their IT infrastructure by providing immediate access to computing resources via the Internet. This change in approach allows organizations to expand their operations effectively while realizing substantial cost reductions. The importance of cloud computing is highlighted by its capability to deliver flexible, scalable, and affordable solutions, enabling companies to respond quickly to market needs without the hassle of large on-site infrastructure. Utilizing cloud services allows companies to improve resource use, boost operational efficiency, and concentrate on primary business goals.

Literature Review

Recent research shows multiple important trends regarding the impact of cloud computing on business functions and information systems, such as:

Scalability: Cloud computing enables companies to adjust resources according to demand, ensuring optimal performance during high-demand times without substantial initial investments.

Cost Effectiveness: The pay as you go model approach of cloud services allows organizations to spend solely on the resources they utilize, lowering capital costs linked to conventional IT infrastructure. Amazon's AWS is a good example of Scalability and Cost-effectiveness.

However, issues such as security risks, the possibility of vendor lock in, and the complexity of overseeing cloud environments have been recognized as potential hazards linked to cloud adoption.

Analysis

Companies are incorporating cloud computing into their activities in multiple ways:

Dynamic Resource Allocation: Businesses leverage cloud platforms to flexibly assign resources, guaranteeing effective management of fluctuating workloads while maintaining performance.

Automated Scaling: Cloud services provide automated scaling features that enable companies to modify computing resources instantaneously based on changing demand.

Disaster Recovery: Cloud computing offers strong disaster recovery options, helping businesses preserve data integrity and accessibility during unexpected incidents.

For example, A newly emerging firm/company relying on cloud network such as AWS, Google Cloud and MS Azure can save upto 40% of the cost using cloud computing rather than buying the hardware and software itself.

Discussion

Although cloud computing presents various benefits, several issues need to be resolved:

Safety: Keeping data outside the premises brings worries about data leaks and access without permission. Establishing strong security measures and compliance procedures is crucial to reduce these risks.

Vendor Lock-In: Relying on one cloud provider may restrict flexibility and raise expenses in the long run. Implementing multi-cloud approaches may assist in alleviating this problem.

Cost Control: In the absence of effective oversight, cloud costs can increase unexpectedly. Executing cost management tactics is essential to uphold budgetary oversight.

Conclusion and Recommendation

Cloud computing greatly improves cost-effectiveness and scalability for companies, allowing them to respond to shifting market trends and maximize resource use. To fully leverage the advantages of cloud adoption, companies ought to:

Create an Extensive Cloud Strategy: Synchronize cloud efforts with business goals for seamless integration.

Invest in Employee Development: Provide personnel with the essential skills to handle and enhance cloud environments efficiently.

Establish Strong Security Practices: Emphasize data safeguarding and adherence to regulations to defend against possible risks.

Upcoming studies should investigate hybrid and multi-cloud approaches, along with the incorporation of new technologies like artificial intelligence and machine learning to improve the functionalities of cloud computing even more.

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